

Sean S. Williams

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EDUCATION

Columbia University | New York, NY

Aug. 2023 - Present

B.S. in Computer Engineering

- **Relevant Coursework:** Advanced Programming in C, Operating Systems, Computer Architecture, Malware Analysis & Reverse Engineering, Databases, Data Structures in Java, Fundamentals of Computer Systems, Digital Systems, Microelectronic Circuits, CS Theory, Discrete Math, Linear Algebra
- **Honors:** Dean's List 2025
- **CS/EE GPA:** 3.88

TECHNICAL SKILLS

Core Languages: C, Python, Rust, x86 Assembly, SQL, Bash

Proficient: TypeScript, JS, CSS, HTML, VHDL, C++, Java

Tools: Git, GitHub, Docker, PostgreSQL, Jupyter, LTspice, AMD Vivado, PowerShell, GDB, IDA Pro, Ghidra

Systems: POSIX, Linux kernel, Linux/BSD/Windows APIs, FPGAs, RF communication, computer architecture

SUMMARY

Third year computer engineering major at Columbia University with a deep fascination for all things computers: from computer graphics and simulation, to cybersecurity research, to OS development and embedded devices. Hands-on experience supporting undergraduate EE labs and contributing to open source projects, as well as developing a 3D narrative-driven video game, full-stack web app, Linux kernel scheduler, and more.

PROFESSIONAL HISTORY

Introduction to Electrical Engineering Course | New York, NY

Jan. 2025 - Present

Undergraduate Teaching Assistant

- Supervise the hands-on laboratory section for Columbia University's Intro to Electrical Engineering course.
- Assist undergraduate students in performing electrical experiments, troubleshooting equipment failures, and collecting and analyzing data using tools such as waveform generators and oscilloscopes.
- Teach electrical engineering concepts and design patterns using practical demonstrations in the lab.
- Grade student lab reports in a timely manner, and cover any points of confusion in person.

Sunday Studios | New York, NY

Apr. 2025 - Present

Programmer

- Developed a narrative-driven 3D video game with a unique art style in an indie startup game studio.
- Saved over 42 hours of compilation time and prevented bugs by implementing a CI/CD pipeline with custom GitHub Actions workflows for the Godot game engine, building for Windows, Linux, and MacOS.
- Maintained a rapidly growing codebase by designing and enforcing code quality standards for GDScript.
- Perfected in-game graphics and optimization using the latest Godot renderer features and best practices.

Outlier AI | Remote

Apr. 2024 - Aug. 2024

Code Specialist

- Improved LLM code generation by contributing clean, correct code and documentation to AI training.
- Attended weekly meetings to connect with project management and learn how to write high-quality code.

PROJECTS

Freezer | Operating Systems

Spring 2026

- Developed a round-robin Linux kernel scheduler with support for SMP and idle load balancing.

Minishell | Advanced Programming

Spring 2025

- Developed a “sh”-inspired shell in pure C with support for nested shells, output piping, and signals.

Book Tracking Application | Intro to Databases

Fall 2025

- Created a web application based on PostgreSQL and Python for user tracking of books and reading status.
- Includes social features such as ratings, global challenges, and optionally public or private book collections.

CV Webpage | Personal Project

Jan. 2026

- Created a personal webpage and blog using Hugo and GitHub Actions for CI/CD hosting on GitHub Pages.

EXTRACURRICULAR EXPERIENCE

Columbia University Gaming Group | New York, NY

Aug. 2023 - Present

Lead Technician

- Manage the technician team of CUGG. Responsible for recruiting technically skilled members to help maintain and protect the club’s equipment, used for game development, streaming, and competitive gaming.
- Personally maintain, upgrade, and keep custody of all of the club’s technology during academic breaks.

Competitive Team Director of the Counter-Strike 2 Esports Division

- Organize tryouts for newcomers, and practice and study sessions for established team members.
- Register subteams for collegiate, Ivy League, and national tournaments based on skill level.
- Review team and individual performance, and study strategies to coach and make roster changes.

CU Cyber | New York, NY

Oct. 2023 - Present

Cybersecurity Club Member

- Participate in club lessons and CTF events, with a focus on binary exploitation and other cybersecurity.
- Learn how to use popular pentesting tools like IDA Pro, Ghidra, GDB, pwntools, Binary Ninja, Burp Suite, etc.